BUSINESS REQUIREMENTS DOCUMENT

AI Tax Document Preparation Assistant

Multi-Agent Tax Document Readiness and Review System

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1. Executive Summary

The AI Tax Document Preparation Assistant is a multi-agent system intended to transform a collection of scattered Indian tax-related documents into a structured, review-ready package. The solution interprets documents, extracts key information, categorizes documents and values, validates the collection for duplicates and inconsistencies, identifies missing or incomplete information, and flags items that require human review. It is designed to assist the user or a qualified tax professional with document readiness and review. It does not make final tax or legal decisions or file tax returns.

The proposed solution uses deterministic processing wherever practical and uses LLM-based reasoning primarily for ambiguous cases. This approach is intended to improve reliability, explainability, latency, and cost efficiency.

2. Problem Solution & Business Case

Tax preparation can involve multiple documents originating from different sources and formats, including Form 16, AIS/TIS, salary slips, investment statements, rent receipts, insurance receipts, invoices, scanned documents, and photographs. Reviewing these documents manually can make it difficult to determine whether the required information is complete, reliable, consistent, and ready for further tax review.

The key problems addressed are:

- Scattered tax-related documents across multiple files and formats.
- Difficulty determining whether extracted information is reliable.
- Difficulty categorizing documents and extracted values into meaningful tax-document groups.
- Duplicate and near-duplicate documents within a collection.
- Missing, incomplete, or inconsistent information across documents.

- Low-confidence extraction that may otherwise go unnoticed.
- Lack of clear source evidence and explanations when an issue is flagged.
- Manual effort required to prepare a clear document-readiness package for review.

3. Objectives & Success Metrics

3.1 Business Objectives

1. Create a structured tax-document readiness package from a collection of uploaded documents.
2. Reduce manual effort involved in organizing and reviewing tax-related documents.
3. Improve visibility into missing, duplicated, incomplete, inconsistent, and low-confidence information.
4. Provide evidence linking extracted information and warnings back to the original document/page.
5. Route ambiguous cases to human review instead of making unsupported assumptions.
6. Improve reliability and cost efficiency by using deterministic methods wherever possible and AI reasoning primarily for ambiguous cases.
7. Provide a concise summary that can be used by the user or a qualified tax professional.

3.2 Key Performance Indicators (KPIs)

≥ 95% **Document processing completion rate**	**≥ 90%** **Extraction Accuracy**	**≥ 90%** **Categorization Accuracy**
≥ 95% **Issue Detection Precision**	**≥ 90%** **Duplicate detection effectiveness**	**≥ 90%** **Evidence Coverage**

4. Scope of Work

4.1 In Scope

- Upload and intake of a collection of tax-related PDFs, images, scans, and photographs.
- Readability assessment and text extraction.
- Extraction of relevant structured information such as dates, amounts, names, document numbers, salary, investments, insurance payments, and rent details.
- Document and value categorization.
- Confidence scoring for extraction and categorization.
- Exact duplicate detection and near-duplicate detection.
- Cross-document and collection-level validation for missing documents, incomplete information, and inconsistencies.
- Low-confidence item identification.
- Evidence and explanation for warnings.
- Adaptive workflow orchestration.
- Human-review routing for ambiguous or uncertain cases.
- Generation of a final tax-document readiness package and one-page summary.

4.2 Out of Scope (Phase 1)

- Making final tax decisions or legal decisions.
- Filing tax returns on behalf of the user.

- Replacing a qualified tax professional.
- Determining which conflicting value is legally correct.
- Making unsupported assumptions when information is ambiguous.

5. Stakeholders and User Personas

Stakeholder / Role	Interest / Responsibility
Taxpayer / End User	Uploads the document collection, provides basic context where required, reviews warnings, and uses the readiness package.
Qualified Tax Professional	Uses the generated package, evidence, warnings, and summary as an aid during tax-document review.
Human Reviewer	Reviews ambiguous classifications, uncertain extractions, and other cases routed for manual attention.
System / AI Agents	Process documents, extract information, categorize content, validate the collection, and orchestrate the workflow.
Project / Business Owner	Defines objectives, approves scope, and evaluates whether the solution meets business needs.

6. Functional Requirements

6.1 Document Intake & Capture

ID	Requirement	Priority
FR-01	System shall allow users to upload tax-related documents in supported PDF and image formats, individually or in bulk.	MUST
FR-02	System shall allow users to provide basic context such as taxpayer information, financial year, assessment year, and document category where required.	MUST
FR-03	System shall detect unsupported, corrupted, password-protected, or unreadable documents and provide a clear error or warning message.	MUST
FR-04	System shall preserve the original uploaded document as source evidence throughout the processing workflow.	MUST
FR-05	System shall assign a unique identifier to each uploaded document and maintain its processing status.	MUST

6.2 Document Processing & OCR

ID	Requirement	Priority
FR-06	System shall preprocess uploaded documents to improve readability and extraction quality where required.	MUST
FR-07	System shall perform OCR on scanned or image-based documents to convert visual content into machine-readable text.	MUST
FR-08	System shall identify document structure and relevant regions before extracting information.	MUST
FR-09	System shall provide confidence indicators for extracted information and identify low-confidence fields for review.	MUST
FR-10	System shall retain a reference to the source document and location from which extracted information was obtained.	MUST

6.3 Information Extraction

ID	Requirement	Priority
FR-11	System shall extract relevant tax and financial information from supported document types.	MUST
FR-12	System shall normalize extracted values such as dates, monetary amounts, taxpayer identifiers, financial years, and document references into a consistent structure.	MUST
FR-13	System shall distinguish between extracted information, inferred information, and information that could not be determined from the available evidence.	MUST
FR-14	System shall associate extracted information with its originating document and provide source evidence for reviewer verification.	MUST
FR-15	System shall allow authorized users to correct extracted information and shall record the correction for audit purposes.	MUST

6.4 Document & Value Categorization

ID	Requirement	Priority
FR-16	System shall classify uploaded documents into relevant tax document categories.	MUST
FR-17	System shall categorize extracted financial information according to the applicable tax preparation categories configured for the solution.	MUST
FR-18	System shall identify documents or information that cannot be confidently categorized and route them for human review.	MUST
FR-19	System should learn from authorized reviewer corrections to improve future categorization accuracy.	SHOULD

6.5 Validation, Duplicate & Inconsistency Detection

ID	Requirement	Priority
FR-20	System shall detect exact and near-duplicate documents within the submitted document collection.	MUST
FR-21	System shall validate extracted information for internal consistency across related documents.	MUST
FR-22	System shall identify missing, incomplete, conflicting, or potentially inconsistent information relevant to tax-document preparation.	MUST
FR-23	System shall identify potential mismatches between taxpayer information appearing across multiple documents.	MUST
FR-24	System shall provide a human-readable explanation for each validation warning or inconsistency identified.	MUST
FR-25	System should prioritize identified issues according to configurable severity levels.	SHOULD

6.6 Human Review and Exception Handling

ID	Requirement	Priority
FR-26	System shall route low-confidence extractions, ambiguous classifications, and validation exceptions to an authorized human reviewer.	MUST
FR-27	Reviewers shall be able to view the extracted information together with the relevant source document or evidence.	MUST
FR-28	Reviewers shall be able to accept, correct, reject, or mark an identified issue as resolved.	MUST

FR-29	System shall record reviewer actions and decisions in the audit trail.	MUST
FR-30	System shall prevent unresolved critical issues from being presented as fully ready for tax-document preparation.	MUST

6.7 Readiness Package & Reporting

ID	Requirement	Priority
FR-31	System shall generate a structured tax-document readiness package summarizing processed documents, extracted information, warnings, and unresolved issues.	MUST
FR-32	System shall provide a missing/incomplete information checklist for the taxpayer or tax professional.	MUST
FR-33	System shall provide source evidence for key extracted information and identified issues.	MUST
FR-34	System shall provide an overall document-readiness status based on configured validation and review rules.	MUST
FR-35	System should allow the readiness package and supporting information to be exported in a review-friendly format.	SHOULD
FR-36	System shall clearly state that the readiness result is an assistive review output and not a final tax or legal determination.	MUST

7. AI/ML Solution Architecture

The solution comprises cooperating document-processing, AI-agent, validation, and human-review components. The architecture automates document preparation while maintaining source evidence, auditability, and clear human decision boundaries.

7.1 Document Processing & OCR Layer

Purpose
Convert uploaded tax documents into structured, machine-readable information while preserving the original source evidence.

- **Document ingestion:** Receives uploaded tax-related documents and assigns document identifiers.
- **Pre-processing:** Improves document quality through operations such as image enhancement, deskewing, and noise reduction where required.
- **Document classification:** Identifies the document type and relevant document category.
- **OCR:** Converts scanned or image-based documents into machine-readable text.
- **Information extraction:** Identifies relevant tax, financial, taxpayer, date, and document-reference information.
- **Confidence scoring:** Assigns confidence indicators to extracted information and identifies low-confidence results for review.

7.2 AI Agent Orchestration Layer

Purpose
Coordinate the end-to-end tax-document readiness workflow using specialized agents operating within defined rules and guardrails.

- **Intake & OCR Agent:** Manages document intake, classification, preprocessing, and OCR processing.
- **Extraction Agent:** Extracts structured information from processed documents and associates it with source evidence.
- **Categorization Agent:** Classifies documents and extracted values into the appropriate tax-preparation categories.
- **Validation Agent:** Checks completeness, consistency, duplicate documents, and cross-document relationships.
- **Orchestrator Agent:** Coordinates the workflow, determines which processing step should execute next, and routes exceptions to human review.
- **Agent guardrails:** Agents operate within defined business rules and are not permitted to make final tax or legal determinations.

7.3 Validation & Review Layer

Purpose
Identify issues that may affect document readiness and provide reviewers with evidence-backed explanations.

- **Completeness validation:** Identifies expected but missing or incomplete information.
- **Duplicate detection:** Identifies exact and near-duplicate documents.
- **Cross-document validation:** Compares related information across submitted documents.
- **Consistency checks:** Identifies conflicting names, identifiers, dates, amounts, or other relevant information.
- **Evidence linking:** Connects warnings and extracted values to the originating document.
- **Human review:** Routes ambiguous or low-confidence cases to an authorized reviewer.

7.4 Decision Boundary

Purpose
Ensure the system supports tax-document preparation without replacing qualified human judgment.

- The system may extract, classify, compare, validate, flag, summarize, and organize information.
- The system may recommend that information requires review based on configured rules or confidence thresholds.
- The system shall not make final tax or legal decisions.
- The system shall not represent uncertain or unsupported information as fact.
- Final tax treatment, interpretation, filing decisions, and professional sign-off remain with the taxpayer or qualified tax professional.

8. Non-Functional Requirements

Category	Requirement
Performance	Document processing and readiness analysis shall complete within an acceptable response time under normal operating conditions.
Scalability	System shall support increasing document volumes without requiring fundamental redesign of the solution architecture.
Availability	System shall provide reliable availability during defined business operating periods.
Security	Tax and personal information shall be protected through encryption, role-based access control, authentication, and secure processing.
Data Privacy	The system shall process and retain taxpayer information in accordance with applicable privacy requirements and organizational policies.
Auditability	Document processing, AI-agent actions, extracted information changes, reviewer decisions, and readiness outcomes shall be traceable.

Explainability	Validation warnings, classifications, and important AI-assisted outputs shall include human-readable explanations and supporting evidence where applicable.
Accuracy	The system shall measure extraction, classification, duplicate detection, and validation performance against an agreed test dataset.
Usability	The interface shall allow taxpayers and qualified tax professionals to review document status, warnings, evidence, and missing information with minimal training.
Reliability	Processing failures shall be handled gracefully without silently losing uploaded documents or previously processed information.
Maintainability	Processing rules, document categories, validation rules, and configurable thresholds should be maintainable without requiring complete system redevelopment.

9. Data Requirements & Governance

9.1 Data Inputs

- **Tax Documents:** Tax-related documents uploaded by the taxpayer or authorized user, including supported financial, income, investment, deduction, and supporting documents.
- **Taxpayer Information:** Basic taxpayer information required to associate documents and perform consistency checks.
- **Financial Information:** Extracted income, deductions, investments, transactions, dates, amounts, and other relevant financial values.
- **Supporting Evidence:** Original documents and source locations associated with extracted information.
- **Reference Data:** Configured document categories, validation rules, expected information, and other business rules required for readiness assessment.

9.2 Data & Evidence Requirements

- Every extracted value should be traceable to the source document from which it was obtained.
- Original documents shall be preserved as source evidence and shall not be overwritten by corrected values.
- Corrections made by authorized reviewers shall be recorded separately from the original extracted value.
- Validation warnings shall retain the information and evidence used to generate the warning.
- The system shall distinguish between source-extracted information, reviewer-corrected information, and AI-generated observations.

9.3 Data Quality & Retention

- The system shall identify incomplete, unreadable, ambiguous, or low-confidence information.
- Duplicate documents shall be identified before being treated as separate evidence.
- Processed records, source documents, extracted information, corrections, and audit logs shall be retained according to organizational policy and applicable requirements.
- Human corrections shall be captured as feedback data where permitted for future model and rule improvement.
- Data retention periods shall be configurable according to applicable organizational and regulatory requirements.

9.4 Security & Privacy

- Access to taxpayer documents and extracted information shall be restricted according to user roles and authorization.
- Sensitive taxpayer information shall be encrypted in transit and at rest.
- Access to documents and extracted information shall be logged.
- The system shall follow data-minimization principles and retain only information required for the defined business purpose.

- AI processing shall operate within approved data-security and privacy boundaries.

10. Integration Requirements

System	Integration Need
Document Storage	Store original uploaded documents, processed documents, extracted information, and supporting evidence securely.
Identity Provider / SSO	Authenticate users and enforce role-based access to taxpayer information and review functions.
OCR / Document Processing Services	Process scanned and image-based documents where OCR capabilities are provided through external or internal services.
Notification Services	Send notifications for processing completion, missing information, validation warnings, and human-review requirements.
Tax / Financial Data Sources	Where available and authorized, retrieve relevant supporting information for validation and cross-document consistency checks.
Reporting / Export Services	Generate review-ready summaries and exportable readiness information.

11. Assumptions, Dependencies & Constraints

11.1 Assumptions

- Users are expected to provide documents relevant to the intended tax-document review.
- Uploaded documents are accessible and in supported formats.
- Document/page evidence can be retained for supported extraction workflows.
- The system is an assistive document-readiness tool and not a replacement for qualified professional judgement.
- Human review is available for cases that the system cannot reliably resolve.
- The exact set of expected documents may depend on the taxpayer/document-set context and must be defined further if automated completeness rules are expanded.

11.2 Dependencies

- Availability and quality of OCR and document-processing capabilities.
- Availability of required reference data and configurable validation rules.
- Availability of approved document storage and identity-management services.
- Availability of suitable test datasets for evaluating extraction, classification, duplicate detection, and validation performance.
- Business and tax-domain stakeholder availability for reviewing ambiguous cases and validating requirements.

11.3 Constraints

- OCR quality may vary based on document image quality.
- LLM reasoning can be uncertain and therefore should not be treated as an authoritative tax/legal decision.
- Completeness checks depend on the document-set context and available rules.
- Cross-document comparisons are limited to information that can be reliably extracted.

- The project document does not define production security, retention, compliance, or deployment specifications.

12. Risks & Mitigation

Risk	Impact	Mitigation
OCR misreads poor-quality or scanned documents	High	Confidence scoring, preprocessing, source-evidence review, and human validation for low-confidence results.
Incorrect document classification	Medium	Classification confidence thresholds, reviewer escalation, and configurable document categories.
Missing or incomplete information is not detected	High	Rule-based completeness checks, configurable expected-document requirements, and reviewer verification.
Conflicting information across documents	High	Cross-document validation, human-readable warnings, and source evidence for each identified conflict.
Duplicate documents are incorrectly identified or missed	Medium	Combination of exact and similarity-based duplicate detection with reviewer confirmation for ambiguous cases.
AI produces unsupported or misleading observations	High	Deterministic checks wherever practical, evidence grounding, confidence thresholds, and explicit decision boundaries.
Sensitive taxpayer information is exposed	High	Encryption, role-based access, least-privilege permissions, secure storage, and audit logging.
Users over-rely on AI-generated results	High	Clear labeling of AI-assisted outputs, source evidence, human-review workflow, and explicit statement that final tax decisions remain with qualified users.
Model performance varies across document formats	Medium	Representative test datasets, confidence-based routing, continuous evaluation, and support for manual correction.
Integration or infrastructure delays	Medium	Early technical discovery, phased implementation, and defined fallback/manual workflows.

13. Timeline and Budget

13.1 Timeline

The project timeline will be finalized during detailed project planning based on the approved scope, technical design, resource availability, and system testing requirements.

The proposed project phases are:

Phase	Key Activities	Timeline
Requirements & Design	Finalize requirements, business rules, architecture, and evaluation criteria	To be finalized
Prototype Development	Develop core document intake, extraction, categorization, and validation capabilities	To be finalized
Integration & Testing	Integrate agents, workflow orchestration, exception handling, and human-review workflows	To be finalized

Evaluation & Validation	Test against a reviewed ground-truth dataset and measure defined KPIs	To be finalized
Deployment Preparation	Finalize security, data handling, deployment, documentation, and operational requirements	To be finalized

13.2 Budget

No final project budget has been established at the requirements stage. The budget shall be determined during technical and project planning based on:

- Development and testing resources
- Infrastructure and storage requirements
- OCR processing costs
- LLM/API usage costs, where applicable
- Security and deployment requirements
- Monitoring and maintenance requirements

The final budget and cost estimates shall be reviewed and approved before production implementation.

14. Acceptance Criteria

- **AC-01:** All "Must" priority functional requirements defined in Section 6 are implemented and pass User Acceptance Testing.
- **AC-02:** Supported tax documents can be uploaded, processed, classified, and associated with their original source evidence.
- **AC-03:** Information extraction meets the agreed accuracy threshold on the approved test dataset.
- **AC-04:** Duplicate and near-duplicate documents are identified according to the agreed evaluation criteria.
- **AC-05:** The system identifies missing/incomplete information and produces a usable checklist for review.
- **AC-06:** Cross-document inconsistencies are identified and presented with human-readable explanations and supporting evidence.
- **AC-07:** Low-confidence and ambiguous cases are successfully routed to human review.
- **AC-08:** Reviewer corrections and decisions are recorded in the audit trail.
- **AC-09:** The system generates a structured tax-document readiness package containing processed documents, extracted information, warnings, unresolved issues, and supporting evidence.
- **AC-10:** Security, privacy, access-control, and auditability requirements are reviewed and approved by the relevant stakeholders.

15. Sign-off

Role	Name	Signature	Date
Business / Project Owner			
Project Supervisor			
Technical Reviewer			
Business Stakeholder			

16. Glossary

Term	Definition
AI Agent	An AI-enabled software component designed to perform a specific task within the tax-document processing workflow.
AI Orchestration	Coordination of multiple AI agents and processing steps to complete the document-readiness workflow.
OCR	Optical Character Recognition: Technology used to convert scanned or image-based documents into machine-readable text.
LLM	Large Language Model: An AI model capable of understanding and generating natural language for reasoning and processing tasks.
RAG	Retrieval-Augmented Generation: an approach that retrieves relevant information from a knowledge source before generating an AI response.
Document Classification	The process of identifying and assigning an uploaded document to an appropriate tax-document category.
Information Extraction	The process of identifying and extracting relevant tax and financial information from documents.
Confidence Score	A measure indicating the system's confidence in an extracted value, classification, or AI-generated result.
Duplicate Detection	The process of identifying documents that are identical or sufficiently similar to previously submitted documents.
Cross-Document Validation	Comparison of information across multiple documents to identify inconsistencies, conflicts, or mismatches.
Human Review	Review performed by an authorized person for ambiguous, low-confidence, or exceptional cases.
Source Evidence	The original document or specific information within a document used to support an extracted value or identified issue.
Readiness Package	A structured collection of processed documents, extracted information, warnings, missing information, and supporting evidence prepared for review.
Decision Boundary	The defined limit between tasks the AI system may perform and decisions that require human judgment.
Ground Truth	A reviewed and validated set of information used as the reference for evaluating system accuracy.
Taxpayer	The individual or entity whose tax-related documents and information are being processed.
Qualified Tax Professional	An authorized tax professional who reviews the readiness package and makes appropriate tax-related judgments.

End of Business Requirements Document